Convex Preferences: What Q3b Is Really Asking

The Setup (from Q3b)

Melsi has preferences: $u(C_1, C_2) = v(C_1) + v(C_2)$
where v is **increasing and concave** (like $v(x) = \ln(x)$ or $v(x) = \sqrt{x}$).
The question asks: "Are her preferences over bundles (C_1, C_2) convex?"

What Does "Convex Preferences" Mean?

Simple Definition:
Convex preferences mean you prefer **averages to extremes**.
If you like bundle A and you like bundle B, then you like the **mixture** (half of A, half of B) at least as much as A or B alone.

In pictures: Your indifference curves are "bowed toward the origin" (convex to the origin).

IMPORTANT: Convex vs Concave Preferences

Don't Get These Confused!

Convex Preferences (what we usually assume):

- You prefer **balanced** bundles (averages)
- Indifference curves are **bowed toward the origin**
- You have **diminishing** marginal rate of substitution
- As you get more of one good, you're willing to give up *less* of the other good to get even more

Concave Preferences (rare and weird):

- You prefer **extreme** bundles (all of one thing)
- Indifference curves are **bowed away from the origin**
- You have **increasing** marginal rate of substitution
- The more you have of one good, the MORE you're willing to give up of the other to get even more
- Example: "The more coffee I drink, the MORE I'm willing to sacrifice donuts for additional coffee" (addiction-like behavior)

Key takeaway: Most goods have **convex preferences** because people like variety and balance. Concave preferences don't make economic sense for typical goods.

Intuitive Example

Suppose you have two bundles:

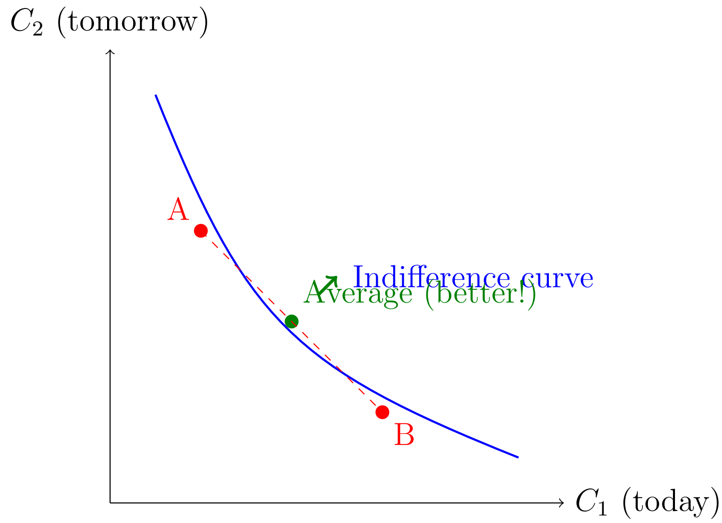
- Bundle A: 10 cookies today, 0 cookies tomorrow
- Bundle B: 0 cookies today, 10 cookies tomorrow

With convex preferences, you'd prefer:

- Bundle C: 5 cookies today, 5 cookies tomorrow (the average)

Why? Because you like balance! Having all your cookies on one day is extreme. Spreading them out is better.

The Graphical Picture



The average bundle is on a **higher** indifference curve than A or B!

What is $\mathrm{MU}_1/\mathrm{MU}_2$?

MU = Marginal Utility

- $MU_1 = \frac{\partial u}{\partial C_1}$ = how much extra utility you get from one more unit of good 1
- $MU_2 = \frac{\partial u}{\partial C_2}$ = how much extra utility you get from one more unit of good 2

The ratio $\mathrm{MU}_1/\mathrm{MU}_2$:
This is your **marginal rate of substitution (MRS)**!

$$MRS = -\frac{dC_2}{dC_1}\Big|_{IC} = \frac{MU_1}{MU_2}$$

It tells you: "How many units of good 2 would I give up for one more unit of good 1?"

In Q3b

For $u(C_1, C_2) = v(C_1) + v(C_2)$:

$$MU_1 = \frac{\partial u}{\partial C_1} = v'(C_1)$$
$$MU_2 = \frac{\partial u}{\partial C_2} = v'(C_2)$$

So:

$$MRS = \frac{MU_1}{MU_2} = \frac{v'(C_1)}{v'(C_2)}$$

The Convexity Test

Preferences are convex if and only if:
As you move along an indifference curve from left to right (increasing C_1, decreasing C_2), the MRS **decreases** (becomes less steep).

In other words: $\frac{MU_1}{MU_2}$ should get smaller as C_1 increases and C_2 decreases.

Why This Makes Sense

Start: You have lots of C_2, very little C_1
⇒ You really want more C_1 (it's scarce!)
⇒ MRS is high (willing to give up lots of C_2 for one unit of C_1)

End: You have lots of C_1, very little C_2
⇒ You already have plenty of C_1, now you want C_2
⇒ MRS is low (not willing to give up much C_2 for more C_1)

This is **diminishing marginal rate of substitution** = convex preferences!

Solving Q3b Step-by-Step

Given: $u(C_1, C_2) = v(C_1) + v(C_2)$ where v is **concave**
Question: Are preferences convex?

Step 1: Find MRS

$$MRS = \frac{MU_1}{MU_2} = \frac{v'(C_1)}{v'(C_2)}$$

Step 2: What happens when $C_1 \uparrow$ and $C_2 \downarrow$?

As we move along an IC:

- C_1 increases: $C_1 \uparrow$
- C_2 decreases: $C_2 \downarrow$

Step 3: Use the fact that v is concave

Concave function means: $v'' < 0$
This means v' is **decreasing**:

- As $C_1 \uparrow$, we have $v'(C_1) \downarrow$ (marginal utility of C_1 falls)
- As $C_2 \downarrow$, we have $v'(C_2) \uparrow$ (marginal utility of C_2 rises)

Step 4: What happens to MRS?

$$MRS = \frac{v'(C_1)}{v'(C_2)}$$

- Numerator $v'(C_1)$ goes down: $v'(C_1) \downarrow$
- Denominator $v'(C_2)$ goes up: $v'(C_2) \uparrow$

Result:

$$\frac{v'(C_1) \downarrow}{v'(C_2) \uparrow} \quad \Rightarrow \quad MRS \downarrow$$

The MRS decreases! This is exactly what we need for convex preferences.

Answer: YES, preferences are convex.

Concrete Example

Let's use $v(x) = \ln(x)$ (which is concave).
So $v'(x) = \frac{1}{x}$ (which is decreasing).

Scenario 1: Lots of C_2, little C_1

Suppose $C_1 = 1$, $C_2 = 10$:

$$v'(C_1) = v'(1) = \frac{1}{1} = 1$$
$$v'(C_2) = v'(10) = \frac{1}{10} = 0.1$$
$$MRS = \frac{1}{0.1} = 10$$

Interpretation: You'd give up 10 units of C_2 for 1 more unit of C_1 (because C_1 is scarce!).

Scenario 2: Lots of C_1, little C_2

Suppose $C_1 = 10$, $C_2 = 1$:

$$v'(C_1) = v'(10) = \frac{1}{10} = 0.1$$
$$v'(C_2) = v'(1) = \frac{1}{1} = 1$$
$$MRS = \frac{0.1}{1} = 0.1$$

Interpretation: Now you'd only give up 0.1 units of C_2 for 1 more unit of C_1 (because C_2 is now scarce!).

The Pattern

$$MRS \text{ went from 10 down to 0.1} \quad \Rightarrow \quad \text{MRS is decreasing} \quad \Rightarrow \quad \text{Convex!}$$

Why Does This Matter?

1. **Convexity ensures interior solutions exist**
   - With convex preferences, you want to balance your consumption
   - You won't go to extremes (all C_1 or all C_2)
2. **Convexity is realistic**
   - Most people prefer variety/balance
   - "Diminishing marginal utility" is a real phenomenon

3. **Mathematically nice**
    - Guarantees unique optimal solution
    - FOC is sufficient (not just necessary)

Summary

Key Concepts:

1. **Convex preferences:** Prefer averages to extremes; ICs are bowed toward origin
2. **MRS = $\mathbf{MU_1/MU_2}$:** The slope of your indifference curve; how willing you are to trade good 2 for good 1
3. **Convexity test:** MRS should decrease as you move along IC (as $C_1 \uparrow$, $C_2 \downarrow$)
4. **For Q3b:**
   - v concave $\Rightarrow$ v' decreasing
   - As $C_1 \uparrow$: $v'(C_1) \downarrow$ (numerator falls)
   - As $C_2 \downarrow$: $v'(C_2) \uparrow$ (denominator rises)
   - Therefore: $\frac{v'(C_1)}{v'(C_2)} \downarrow$ (MRS falls)
   - Conclusion: Preferences are convex!

The Connection

Concave v function
$\Downarrow$
Marginal utility v' is decreasing
$\Downarrow$
MRS $= \frac{v'(C_1)}{v'(C_2)}$ decreases along IC
$\Downarrow$
Convex preferences
$\Downarrow$
You prefer balanced bundles to extreme bundles

Common Mistakes to Avoid

- **Don't confuse:** "concave function" vs "convex preferences"
  - In Q3b: v is **concave**, but preferences are **convex**!
  - Concave v $\Rightarrow$ diminishing marginal utility $\Rightarrow$ convex preferences
- **Remember the direction:** When checking convexity, you need MRS to **decrease** as C_1 increases
- **Be clear about notation:** MRS can be written as:
  - $MRS = -\frac{dC_2}{dC_1}$ (negative slope of IC)
  - $|MRS| = \frac{MU_1}{MU_2}$ (absolute value)

Sometimes people drop the negative sign for simplicity.